Estimations

1 For each of the following expressions:

i Round both numbers to the nearest 10. **ii** Calculate the estimate.

a $99 + 61$ **b** $63 - 22$ **c** 28×31 **d** 447×21

2 For each of the following expressions:

i Round the numbers to the nearest 100, where appropriate.

ii Calculate the estimate.

a $898 - 132$ **b** $765 + 171$ **c** 347×5

3 For each of the following expressions:

i Round both numbers to the nearest 1000.

ii Calculate the estimate.

a $7452 + 4193$ **b** $9848 - 3469$

4 Estimate the value of the following:

a $2692 + 3669$ **b** $3620 - 1082$

c $6534 + 7418 + 1240$ **d** 446×54

e 448×39 **f** $8163 \div 23$

g $4127 \div 19$ **h** $5523 + 321 + 59$

5 Which of the following is closest to the sum $33 + 42 + 24$?

A 10 **B** 100 **C** 1000

6 Consider the sum $947 + 826 + 12$.

a Estimate the answer if each number is rounded to the nearest ten.

b Estimate the answer if each number is rounded to the nearest hundred.

c Find the exact value of the sum.

d Which answer is the most accurate, part (a) or part (b)?

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7 Consider the sum $24.6 + 718.94$.

- **a** Estimate the sum by rounding each number to the nearest ten.
- **b** Estimate the sum by rounding each number to the nearest hundred.
- **c** Find the exact value of the sum.
- **d** Which estimate is the most accurate, part (a) or part (b)?

8 For each of the following multiplication problems:

- **i** Estimate the answer to the multiplication.
- **ii** Check your estimate in part (a) is close, by calculating the actual product, rounded to the nearest tenth.

a 9.82×12.98 **b** 19.66×7.12 **c** 94.12×0.97 **d** 2.93×995

9 For each of the following division problems:

- **i** Estimate the answer to the division.
- **ii** Check your estimate in part (a) is close, by calculating the actual division, rounded to the nearest tenth.

a $26.99 \div 8.98$ **b** $41.72 \div 2.86$ **c** $54.12 \div 0.89$ **d** $182.33 \div 5.98$

Checking reasonableness

10 Are the following calculations correct?

a $57 - 28 = 31$
b $7928 + 14 = 7942$
c $9352 + 47 = 479\,352$
d $4192 + 5968 = 9160$
e $9533 - 5933 = 4400$
f $189\,672 \div 14 = 189\,672 \div 2 \div 7$

11 Are the following statements reasonable?

- **a** $59 - 27$ is approximately equal to 30.
- **b** $68 - 27$ is approximately equal to 60.
- **c** $2883 - 2831$ is approximately equal to $80 - 30$.
- **d** $6871 - 5832$ is approximately equal to $70 - 30$.
- **e** $67 - 29$ is approximately equal to 40.
- **f** 4×88 will be less than 320.
- **g** 4×83 will be greater than 320.

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h $3024 \times 5 = 151\,020$

i 9352×4761 will be greater than $36\,000\,000$.

12 Suppose we want to check the subtraction problem $82 - 80 = 2$.

a Which two numbers must we add?

b What should the two numbers add up to?

13 State whether the following products are close to 1819:

a 98×97 **b** 201×9 **c** 33×59 **d** 33×18

14 Is the following division close to 200?

$$624 \div 3$$

15 State whether the following is true or false:

a Dividing by 6 is the same as dividing by 3 and then dividing by 3 again.

b 4000 is a reasonable estimate for 42×8.

c 400 is a reasonable estimate for $2002 \div 5$.

d If apples cost $\$3.25/\text{kg}$, then 3.5 kg of apples will cost about $\$9.25$.

e If oranges cost $\$2.25/\text{kg}$, then 5 kg of oranges will cost about $\$11$.

Applications

16 Weatley Coach Lines buses will carry on average 33 passengers per trip. Each day, the company sends out 78 buses and each bus completes on average 23 trips per day. Estimate the number of people that travel on the buses each day.

17 Fiona started a savings account with the aim of reaching $\$272\,088$. If, on average, Fiona deposits $\$5200$ every month, estimate the number of months it will take to reach her goal.

18 Ursula collects beetles. She knows that she has exactly 169 of them. Ursula has to move house this week, and must buy storage boxes to transport them.
Will 9 boxes that hold 20 beetles each be enough storage?

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19 Brad and Bill went out one evening. Bill said:
"I have $\$18.40$, which is about $\$20$. This food stall charges $\$9.50$ for a kebab, which is about $\$10$. That means I have enough money to pay for both of us to get a kebab."

a Is Bill being reasonable? Explain your answer.

b Brad says, "I can put $\$5$ towards our meal, then we can both afford a kebab." Is Brad correct? Explain your answer.

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